Game Project

Rubik’s Cube Simulator

Train and challenge players from all over the world!

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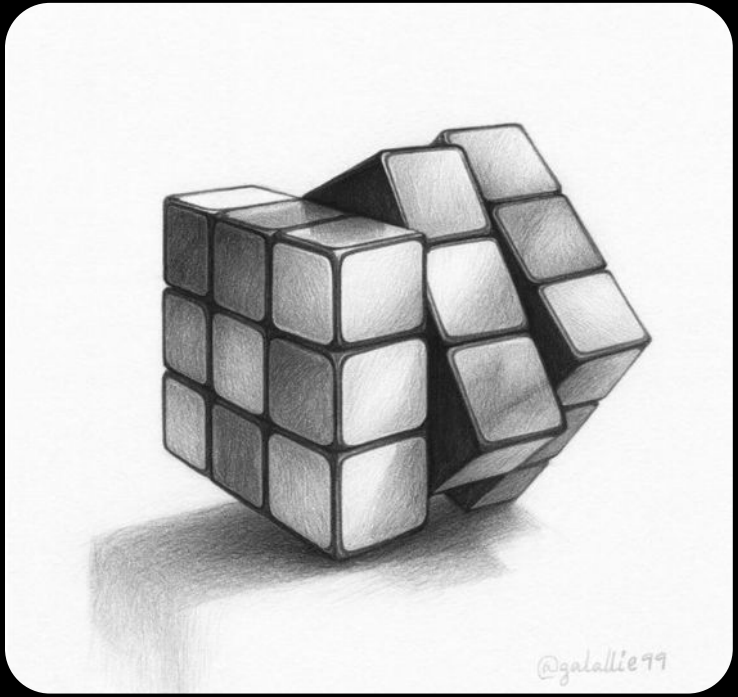

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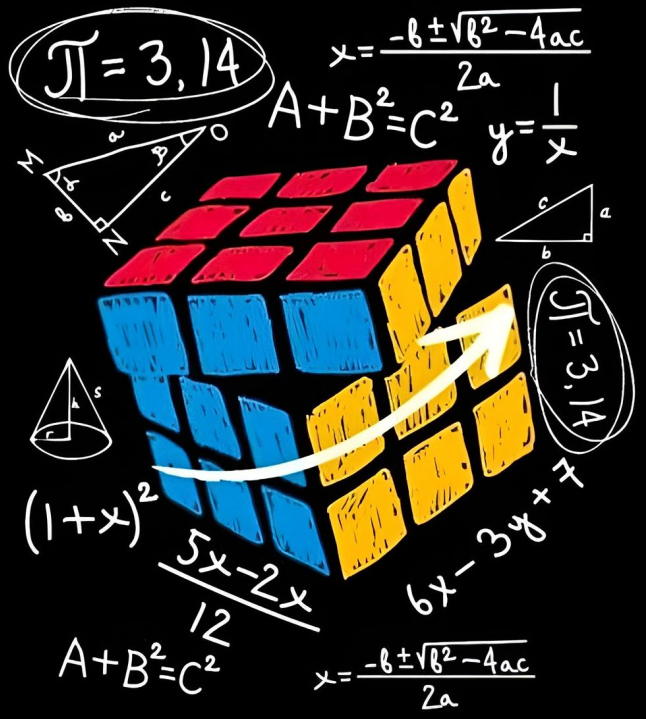
Logic and Entertainment

Our Idea

The core goal of Rubik's Cube Simulator is to combine the physical and tactile appeal of the puzzle with the convenience of the digital world.

Many digital cubes feel clunky or purely decorative; we wanted to create a true "digital simulator" immersed in a high-quality 3D environment.

It's not just a digital game, but a space dedicated to personal and collective growth, designed to ease newcomers into the world of cubing and guide every new enthusiast step by step.



What do we offer?

A realistic 3D simulation with smooth movements and immediate feedback.

How are we doing it?

Through a complete, easy and intuitive application designed to deliver an optimal user experience.

Project Goals

Who are we?

Two students developers passionate about logic puzzle and videogames.

Why are we doing this?

To make the puzzle accessible to everyone and create a platform to learn and compete.

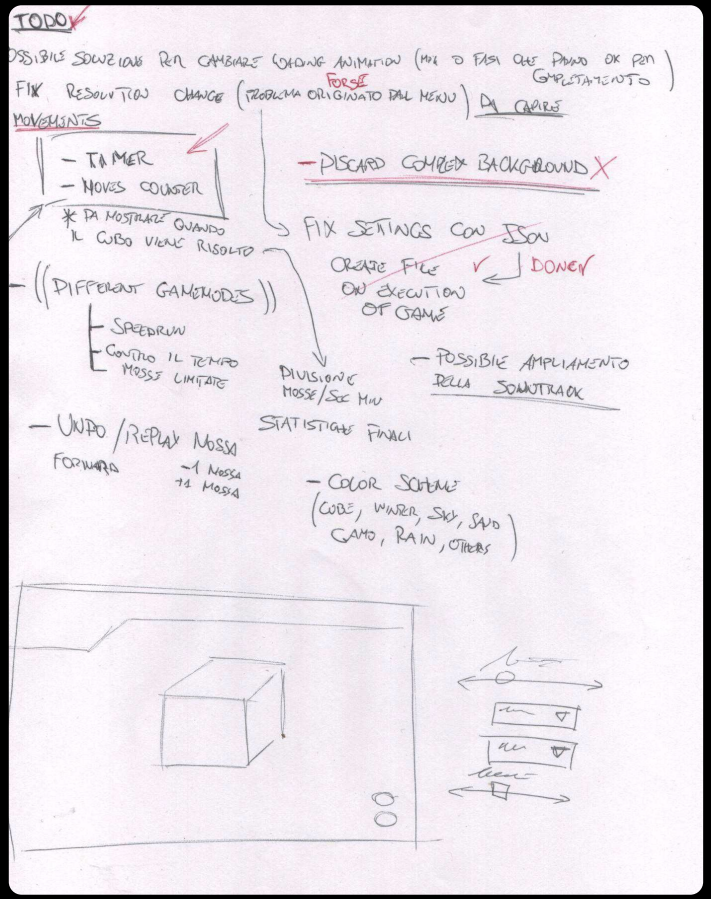
Design and Planning

Design

The simulator was developed entirely in Python, chosen for its flexibility and the richness of its available library ecosystem.

The project is built on two main technical components:

- PyGame, used for managing the game loop, keyboard and mouse input, the user interface, and the overall application logic.
- PyOpenGL, which enabled the implementation of a 3D rendering engine capable of representing the cube with full visual and mechanical accuracy.



Skills

The project was conceived and developed entirely independently, without any external support from teachers.

Every design decision, from the mechanics logic to the technical implementation of the code, is the result of original and independent work.

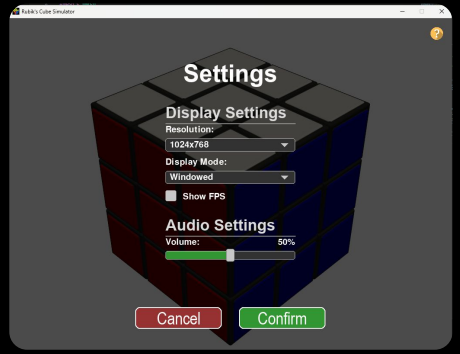
The soundtrack is also an entirely original contribution, composed and produced from scratch by Gianluca Soban to give the game a unique sonic identity consistent with the gameplay experience.

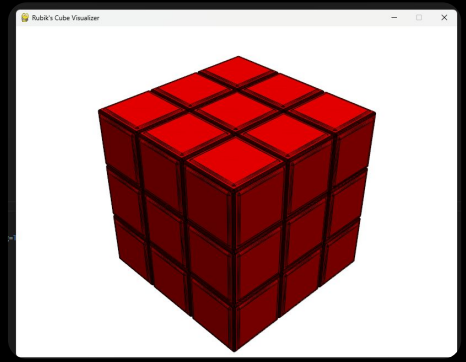
Genesis

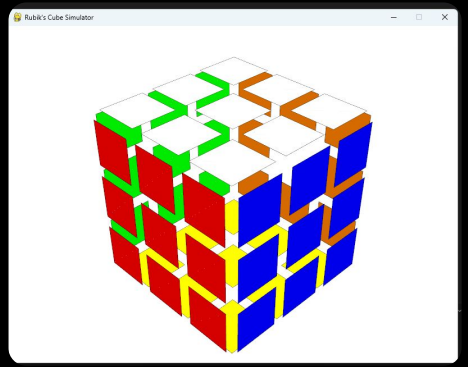
The project originated from essential prototypes focused exclusively on the complex logical management of the cube in a 3D environment.

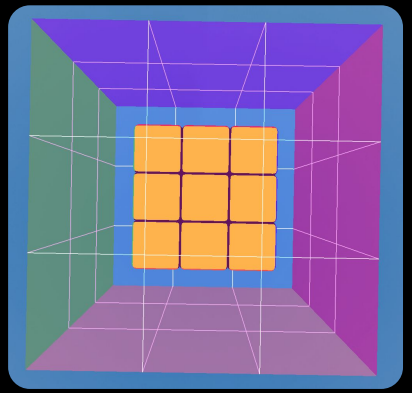
Each development phase introduced new levels of complexity: from the implementation of game modes and the hint system, to the integration of leaderboards and objectives.

This journey documents significant technical growth, in which every structural challenge was transformed into an opportunity to optimize the code and build a solid, complete gameplay experience.





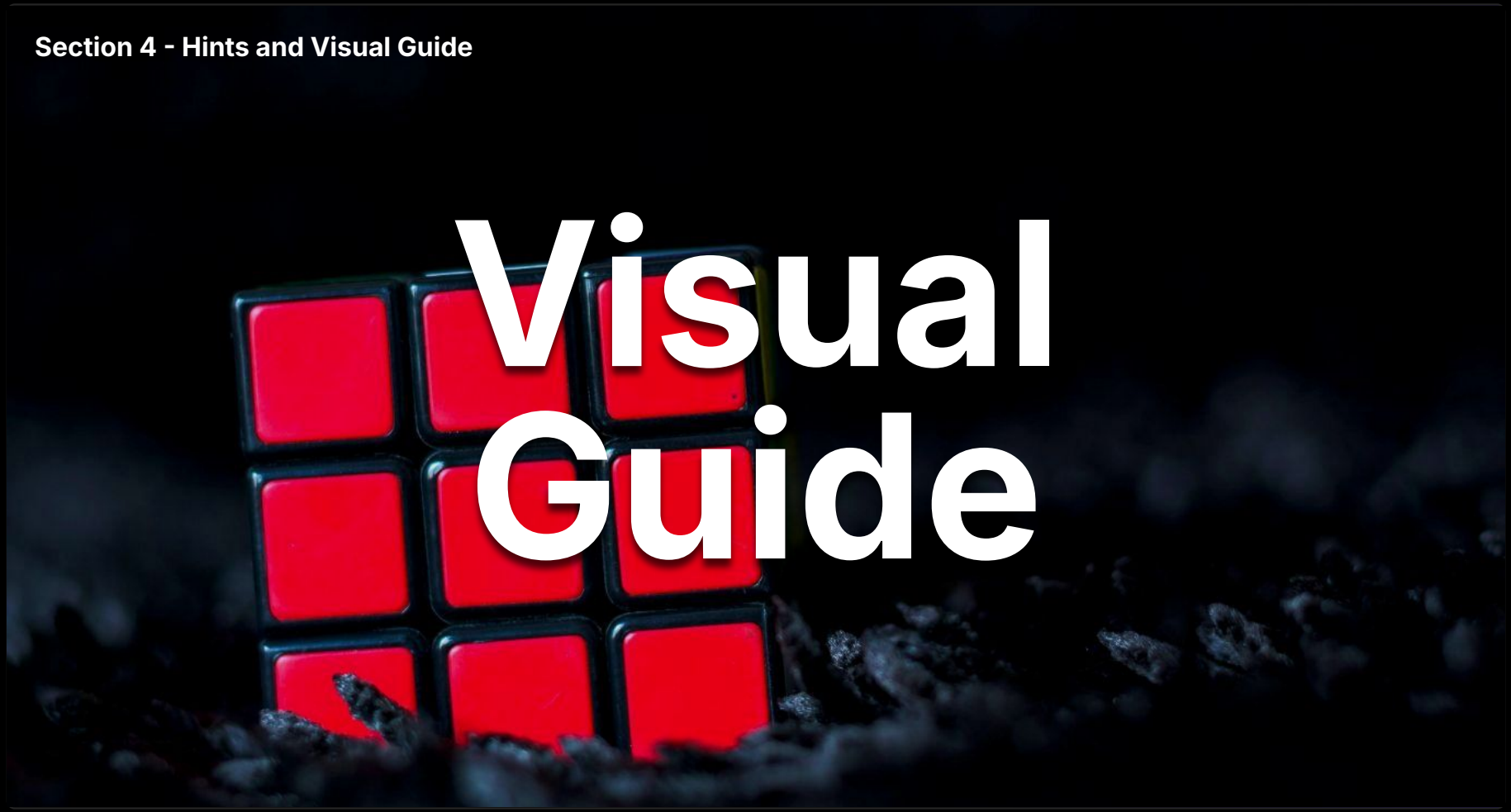




Game Modes

Game Modes

- **Free play:** Solve at you own pace, no time limits.
- **Easy Mode:** 5-move scramble, perfect for learning the basics and getting familiar with the cube.
- **Medium Mode:** 10-move scramble, ideal for those who want to practice more advanced techniques.
- **Hard Mode:** Fully randomized scramble, designed for experienced solvers who want to push to the limit.
- **Limited time Mode:** Solve the cube before the time runs out.
- **Limited moves Mode:** Solve the cube with a fixed number of moves, testing efficiency and planning.
- **Daily Mode:** A new scramble every day for a continuous challenge and steady growth.



Visual Guide

Visual Guide

The visual guide system is designed to be helpful without being intrusive.

Based on an internal solver capable of finding instant solutions, the system adapts hints to the chosen difficulty level, keeping the experience always educational:

- **Easy mode**: 1 hint
- **Medium mode**: 2 hints
- **Hard mode**: 5 hints

This way the simulator becomes a true tutor that teaches you how to solve the cube, not an automatic solution that takes away the thrill of the challenge.

Statistics

Personal Records

Track your best times and move counts for each mode.

Welcome to your personal archive.

More than a simple list of results, this space tells the story of your evolution as a cuber.

Here you'll find in detail your record times, most efficient solutions, and total completed solves, all broken down by difficulty level.

Global Leaderboard

Discover where you rank among the world's best solvers.

The ultimate hall of fame.

It brings together the platform's top players based on their record times across all modes.

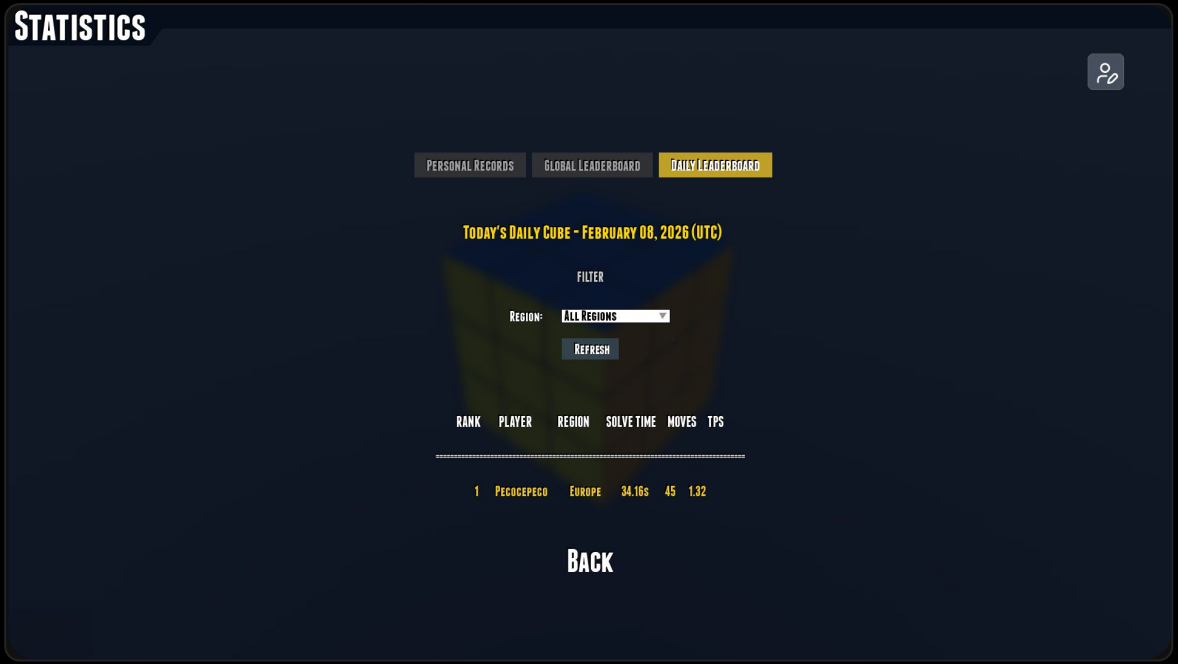
A constant motivation to perfect your technique and memorize algorithms.

Daily Leaderboard

To keep the community active every day, we offer a Universal Daily Scramble that resets for everyone at 00:00 UTC.

All players around the world face the exact same cube configuration for the entire UTC day.

The Daily Leaderboard rewards whoever solves it most efficiently, becoming a daily fixture that every cuber can look forward to each morning, wherever they are in the world.



Achievements

Achievements

Achievements are structured milestones that capture every phase of a cuber's growth.

Each one represents a significant moment in the journey: from the very first solve to mastering the cube under the most difficult conditions.

The system rewards six different skill categories: speed, efficiency, consistency, and the ability to perform under pressure.

The next milestone is always just around the corner.

Credits

This project was created by:

Leonardo Malannino
&
Gianluca Soban

Class 5IA 2025-2026